Algorithms

Graph Searching Techniques
Topological Sorting

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Graph Searching

- Given: a graph G = (V, E), directed or undirected
- Goal: methodically explore every vertex and every edge
- Ultimately: build a tree on the graph
  - Pick a vertex as the root
  - Choose certain edges to produce a tree
  - Note: might also build a *forest* if graph is not connected

- There are two standard graph traversal techniques:
  - Breadth-First Search (BFS)
  - Depth-First Search (DFS)

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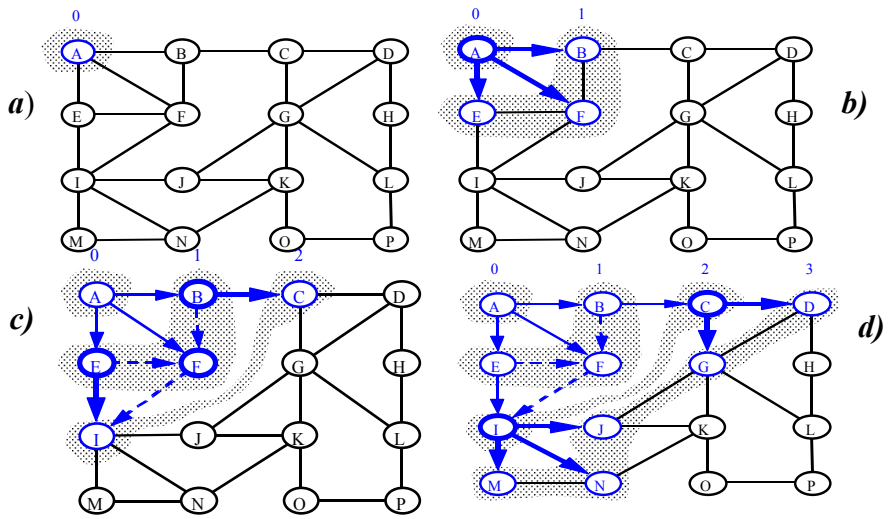
Breadth-First Search

- "Explore" a graph, turning it into a tree
  - One vertex at a time
  - Expand frontier of explored vertices across the *breadth* of the frontier
- Builds a tree over the graph
  - Pick a *source vertex* to be the root
  - Find ("discover") its children, then their children, etc.

Breadth-First Search

- Again will associate vertex "colors" to guide the algorithm
  - White vertices have not been discovered
    - All vertices start out white
  - Grey vertices are discovered but not fully explored
    - They may be adjacent to white vertices
  - Black vertices are discovered and fully explored
    - They are adjacent only to black and grey vertices
- Explore vertices by scanning adjacency list of grey vertices

BFS - A Graphical Representation



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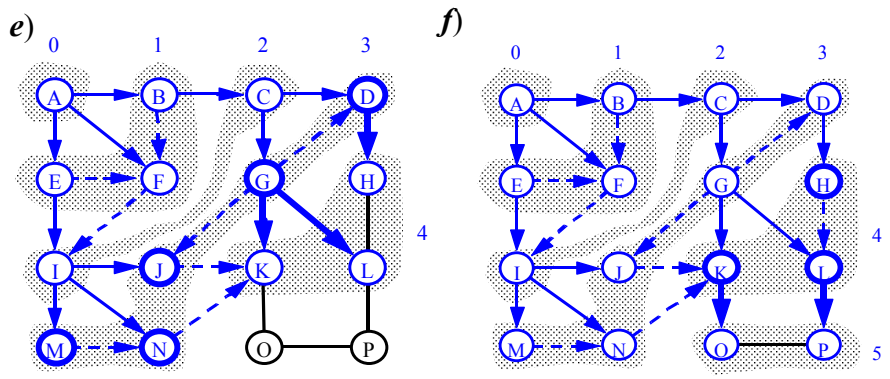
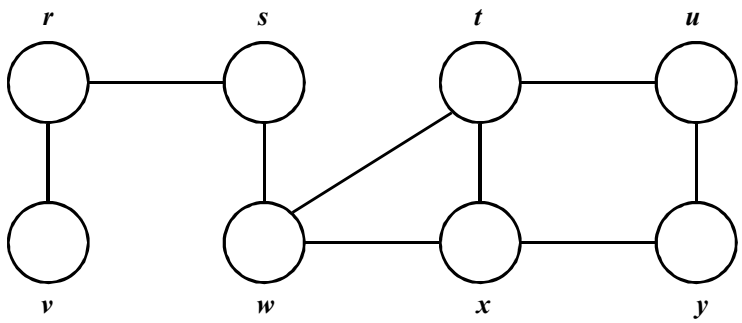


Breadth-First Search: Example



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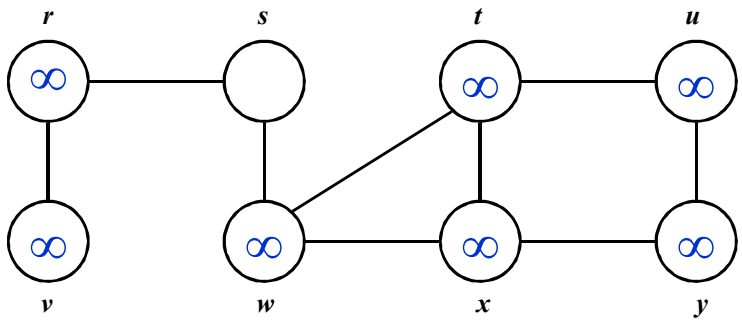
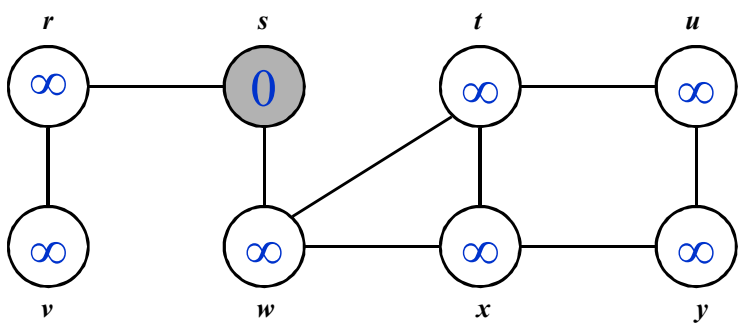
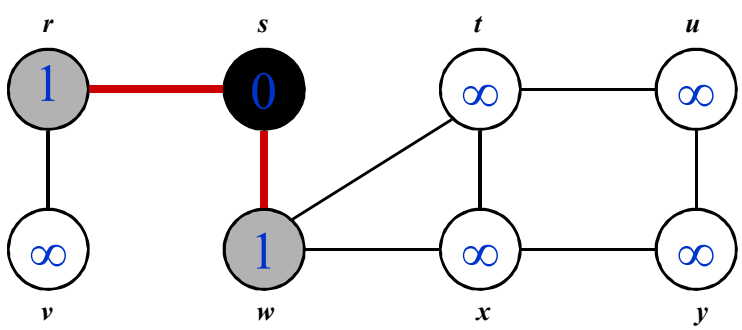


Breadth-First Search: Example



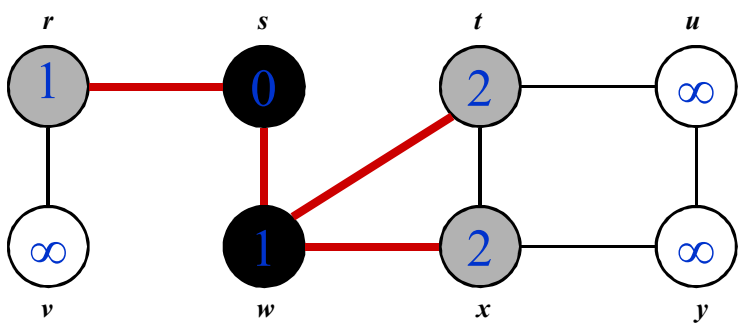
Q: | *s* |

Breadth-First Search: Example



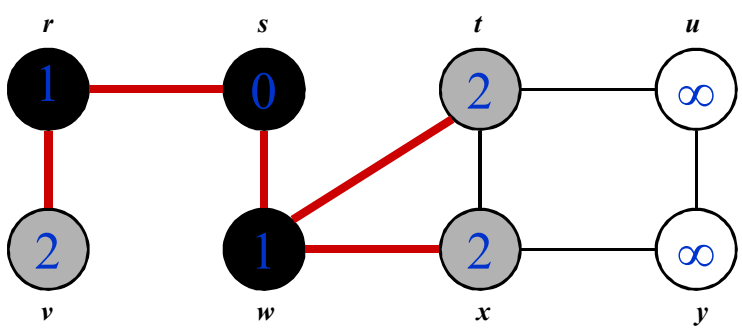
Q: | *w* | *r* |

Breadth-First Search: Example



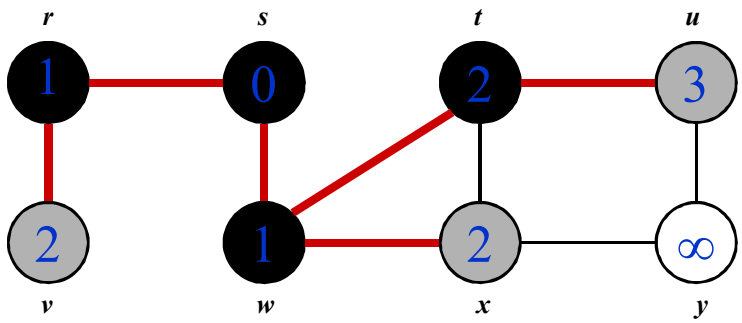
Q: | *r* | *t* | *x* |

Breadth-First Search: Example



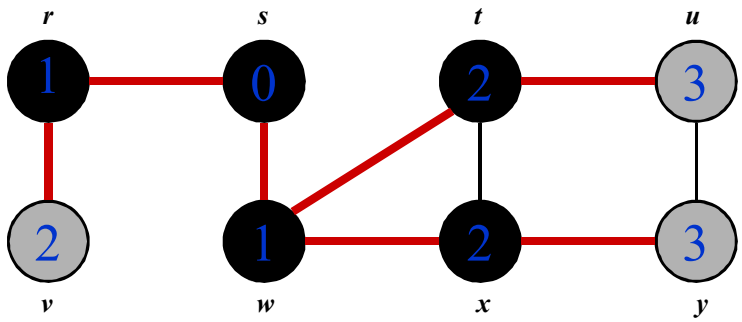
Q: | *t* | *x* | *v* |

Breadth-First Search: Example



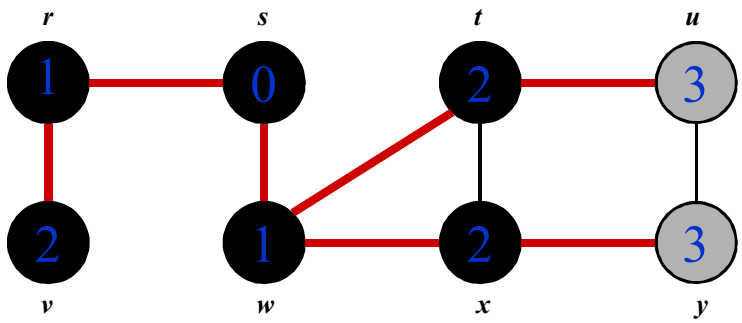
Q: | *x* | *v* | *u* |

Breadth-First Search: Example



Q: | *v* | *u* | *y* |

Breadth-First Search: Example



Q: | *u* | *y* |

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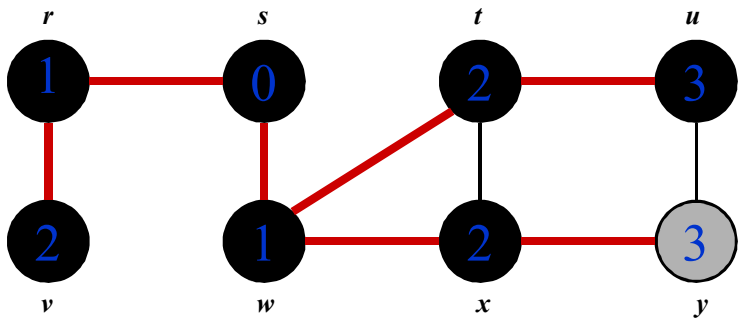

Q: | *y* |

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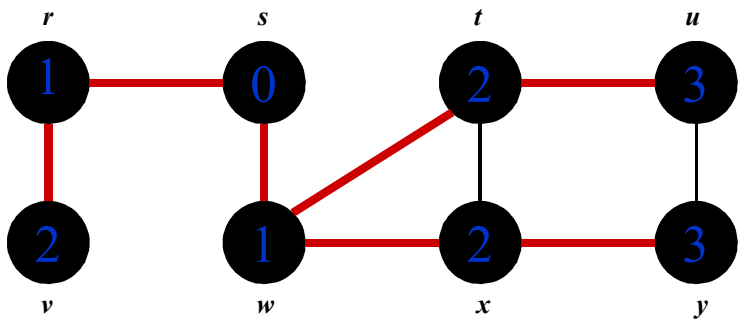
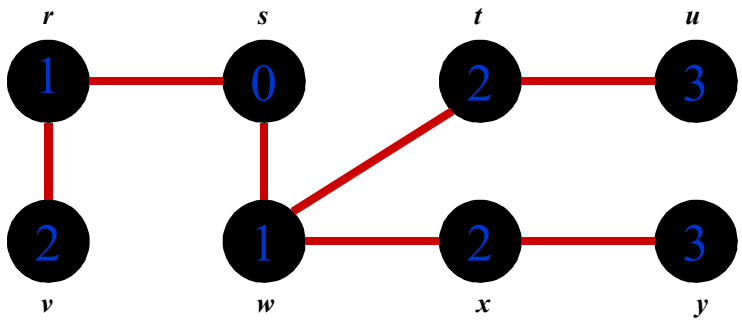


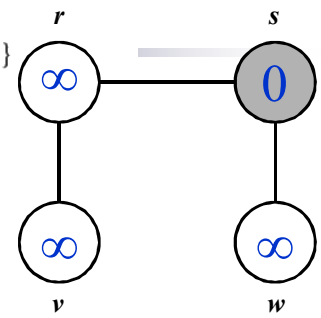
Q: Ø

Breadth-First Search: Example



Output: BFS Spanning Tree

```
BFS(G, s)
for each vertex u ∈ V[G] − {s}
    do color[u] ← WHITE
       d[u] ← ∞
       π[u] ← NIL
color[s] ← GRAY
d[s] ← 0
π[s] ← NIL
Q ← ∅
ENQUEUE(Q, s)
while Q ≠ ∅
    do u ← DEQUEUE(Q)
       for each v ∈ Adj[u]
           do if color[v] = WHITE
                 then color[v] ← GRAY
                      d[v] ← d[u] + 1
                      π[v] ← u
                      ENQUEUE(Q, v)
       color[u] ← BLACK
```



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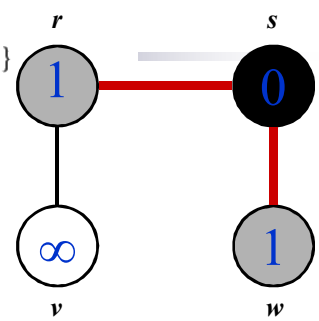
       color[u] ← BLACK
```



Q: | w | r |

BFS: The Code Again

```
BFS(G, s) {
    initialize vertices;
    Q = {s};
    while (Q not empty) {
        u = RemoveTop(Q);
        for each v ∈ u->adj {
            if (v->color == WHITE)
                v->color = GREY;
                v->d = u->d + 1;
                v->p = u;
                Enqueue(Q, v);
        }
        u->color = BLACK;
    }
}
```

Touch every vertex: O(V)

u = every vertex, but only once (Why?)

So v = every vertex that appears in some other vert's adjacency list

What will be the running time?

Total running time:
$O(V + \Sigma(\text{degree}(v))) = O(V+E)$

BFS: The Code Again

```
BFS(G, s) {
    initialize vertices;
    Q = {s};
    while (Q not empty) {
        u = RemoveTop(Q);
        for each v ∈ u->adj {
            if (v->color == WHITE)
                v->color = GREY;
                v->d = u->d + 1;
                v->p = u;
                Enqueue(Q, v);
        }
        u->color = BLACK;
    }
}
```

What will be the storage cost in addition to storing the tree?

Total space used:
$O(V + \Sigma(\text{degree}(v))) = O(V + E)$

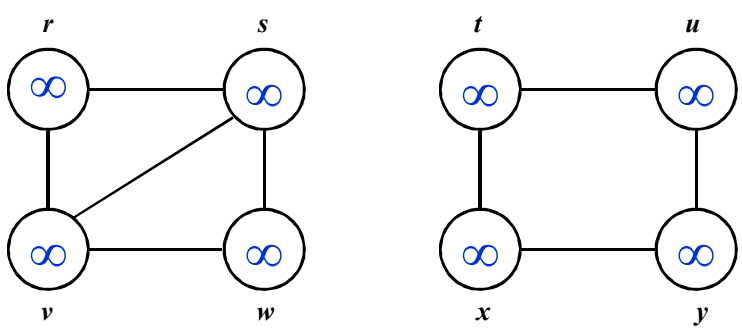
Breadth-First Search : Disconnected Graph

```
BFS(G)
  for each vertex u ∈ G->V
      u->color = WHITE;
      u->d = ∝
      u->π = NIL

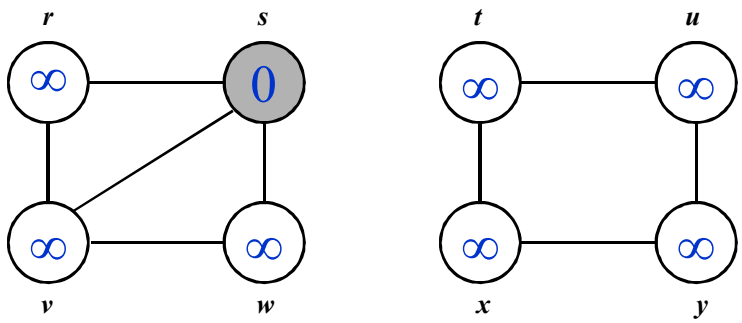
  for each vertex u ∈ G->V
      if (u->color == WHITE)
          BFS_Visit(u);
```

```
BFS_Visit(u)
  u->color = GREY;
  u->d = 0
  u->π = NIL
  Q = ϕ;
  Enqueue(Q, u)
  while Q ≠ ϕ
      u = Dequeue(Q)
      for each v ∈ u->Adj[ ]
          if (v->color == WHITE)
              v->color = GREY;
              v->d = u->d + 1
              v->π = u
              Enqueue(Q, v)
      u->color = BLACK;
```


Breadth-First Search : Disconnected Graph

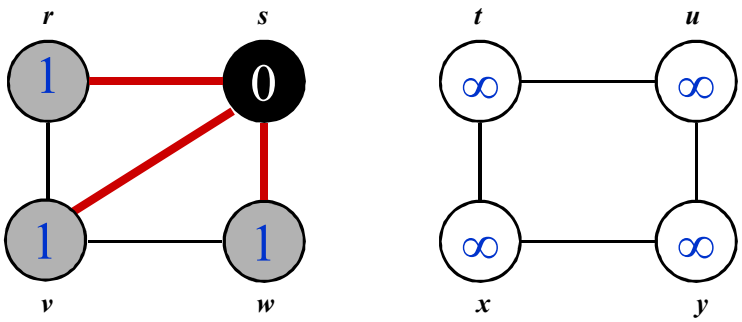


Breadth-First Search : Disconnected Graph



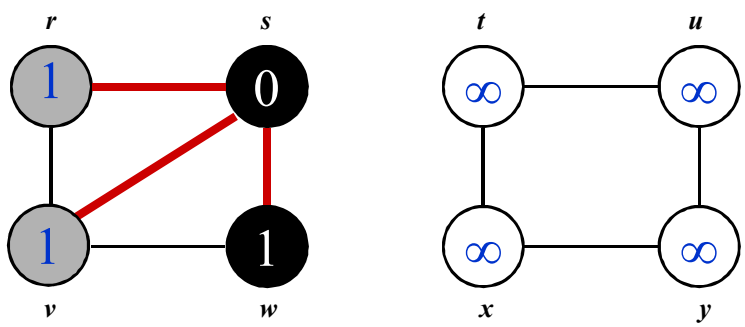
Q: | *s* |

Breadth-First Search : Disconnected Graph



Q: | *w* | *r* | *v* |

Breadth-First Search : Disconnected Graph



Q: | *r* | *v* |

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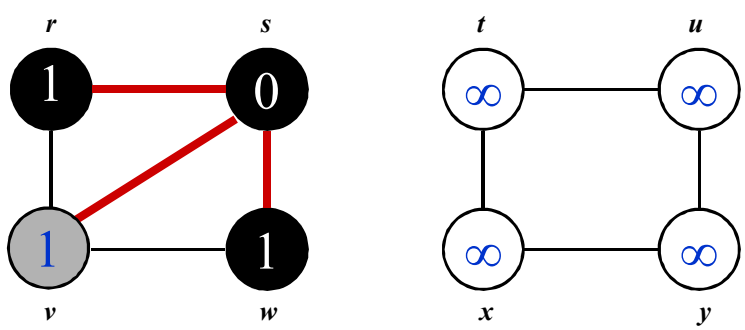
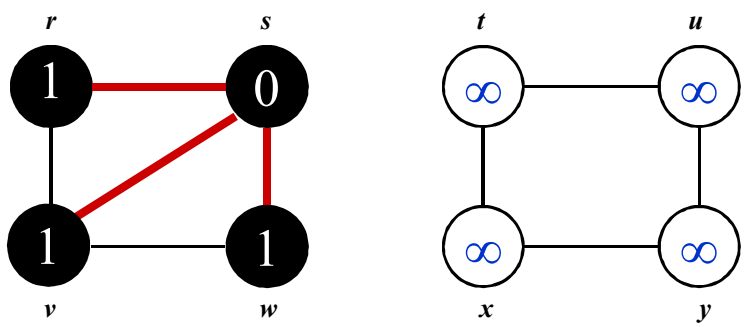


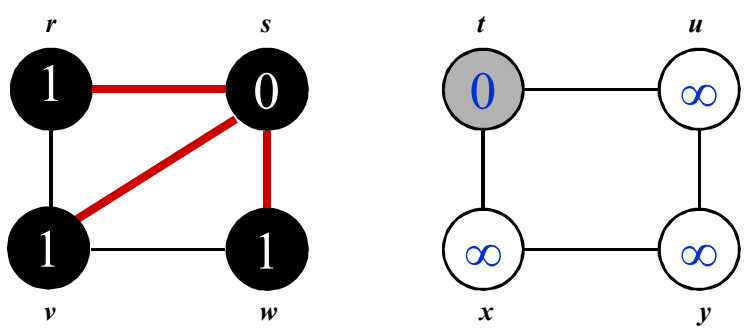
Q: | *v* |

Breadth-First Search : Disconnected Graph



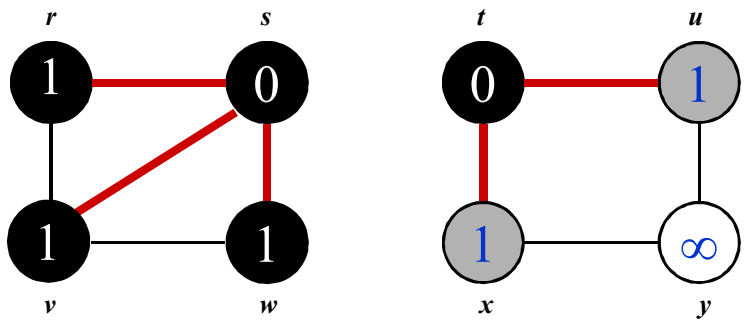
Q: Ø

Breadth-First Search : Disconnected Graph



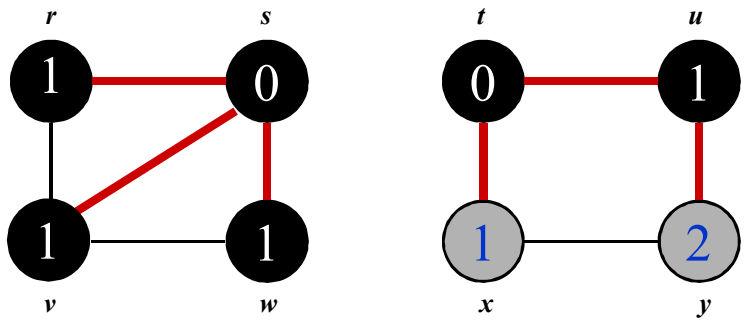
Q: $\boxed{t}$

Breadth-First Search : Disconnected Graph



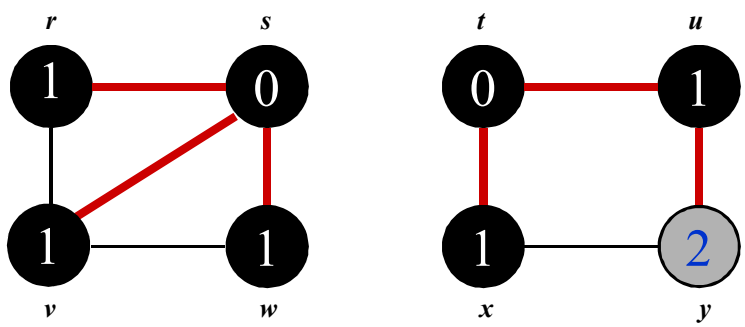
Q: | *u* | *x* |

Breadth-First Search : Disconnected Graph



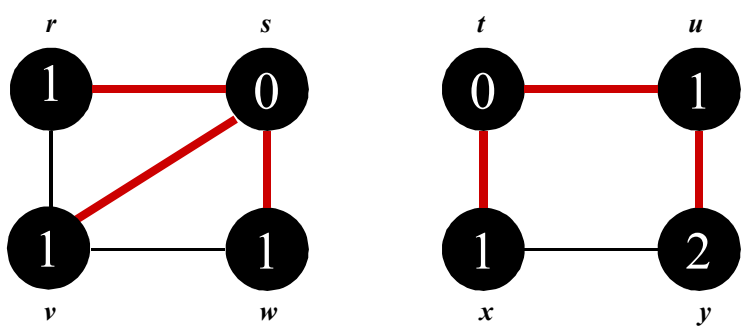
Q: | *x* | *y* |

Breadth-First Search : Disconnected Graph



Q: $\boxed{y}$

Breadth-First Search : Disconnected Graph



Q: Ø

Breadth-First Search: Properties

- BFS calculates the *shortest-path distance* to the source node
  - Shortest-path distance $\delta(s, v)$ = minimum number of edges from s to v, or ∞ if v not reachable from s
- BFS builds *breadth-first spanning tree* (*forest*), in which paths to root(s) represent shortest paths in G
  - Thus can use BFS to calculate shortest path from one vertex to another in $O(V + E)$ time in an unweighted graph

Depth-First Search

- *Depth-first search* is another strategy for exploring a graph
  - Explore "deeper" in the graph whenever possible
  - Edges are explored out of the most recently discovered vertex v that still has unexplored edges
  - When all of v's edges have been explored, backtrack to the vertex from which v was discovered
- Vertices initially colored white
- Then colored grey when discovered
- Then black when finished

Depth-First Search: The Code

```
DFS(G)
{
    for each vertex u ∈ G->V
    {
        u->color = WHITE;
    }
    time = 0;
    for each vertex u ∈ G->V
    {
        if (u->color == WHITE)
            DFS_Visit(u);
    }
}
```

```
DFS_Visit(u)
{
    u->color = GREY;
    time = time+1;
    u->d = time;
    for each v ∈ u->Adj[]
    {
        if (v->color == WHITE)
            DFS_Visit(v);
    }
    u->color = BLACK;
    time = time+1;
    u->f = time;
}
```


Depth-First Search: The Code

```
DFS(G)
{
    for each vertex u ∈ G->V
    {
        u->color = WHITE;
    }
    time = 0;
    for each vertex u ∈ G->V
    {
        if (u->color == WHITE)
            DFS_Visit(u);
    }
}
```

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DFS_Visit(u)
{
    u->color = GREY;
    time = time+1;
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    for each v ∈ u->Adj[]
    {
        if (v->color == WHITE)
            DFS_Visit(v);
    }
    u->color = BLACK;
    time = time+1;
    u->f = time;
}
```

What does `u->d` represent?

Depth-First Search: The Code

```
DFS(G)
{
    for each vertex u ∈ G->V
    {
        u->color = WHITE;
    }
    time = 0;
    for each vertex u ∈ G->V
    {
        if (u->color == WHITE)
            DFS_Visit(u);
    }
}
```

```
DFS_Visit(u)
{
    u->color = GREY;
    time = time+1;
    u->d = time;
    for each v ∈ u->Adj[]
    {
        if (v->color == WHITE)
            DFS_Visit(v);
    }
    u->color = BLACK;
    time = time+1;
    u->f = time;
}
```

What does `u->f` represent?

Depth-First Search: The Code

```
DFS(G)
{
    for each vertex u ∈ G->V
    {
        u->color = WHITE;
    }
    time = 0;
    for each vertex u ∈ G->V
    {
        if (u->color == WHITE)
            DFS_Visit(u);
    }
}
```

```
DFS_Visit(u)
{
    u->color = GREY;
    time = time+1;
    u->d = time;
    for each v ∈ u->Adj[]
    {
        if (v->color == WHITE)
            DFS_Visit(v);
    }
    u->color = BLACK;
    time = time+1;
    u->f = time;
}
```

Will all vertices eventually be colored black?

Depth-First Search: The Code

```
DFS(G)
{
    for each vertex u ∈ G->V
    {
        u->color = WHITE;
    }
    time = 0;
    for each vertex u ∈ G->V
    {
        if (u->color == WHITE)
            DFS_Visit(u);
    }
}
```

```
DFS_Visit(u)
{
    u->color = GREY;
    time = time+1;
    u->d = time;
    for each v ∈ u->Adj[]
    {
        if (v->color == WHITE)
            DFS_Visit(v);
    }
    u->color = BLACK;
    time = time+1;
    u->f = time;
}
```

What will be the running time?

Depth-First Search: The Code

```
DFS(G)
{
    for each vertex u ∈ G->V
    {
        u->color = WHITE;
    }
    time = 0;
    for each vertex u ∈ G->V
    {
        if (u->color == WHITE)
            DFS_Visit(u);
    }
}
```

```
DFS_Visit(u)
{
    u->color = GREY;
    time = time+1;
    u->d = time;
    for each v ∈ u->Adj[]
    {
        if (v->color == WHITE)
            DFS_Visit(v);
    }
    u->color = BLACK;
    time = time+1;
    u->f = time;
}
```

Running time: $O(n^2)$ because call DFS_Visit on each vertex, and the loop over Adj[] can run as many as $|V|$ times

Depth-First Search: The Code

```
DFS(G)
{
    for each vertex u ∈ G->V
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        if (u->color == WHITE)
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    }
}
```

```
DFS_Visit(u)
{
    u->color = GREY;
    time = time+1;
    u->d = time;
    for each v ∈ u->Adj[]
    {
        if (v->color == WHITE)
            DFS_Visit(v);
    }
    u->color = BLACK;
    time = time+1;
    u->f = time;
}
```

BUT, there is actually a tighter bound.

How many times will DFS_Visit() actually be called?

Depth-First Search: The Code

```
DFS(G)
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    for each vertex u ∈ G->V
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    }
}
```

```
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    for each v ∈ u->Adj[]
    {
        if (v->color == WHITE)
            DFS_Visit(v);
    }
    u->color = BLACK;
    time = time+1;
    u->f = time;
}
```
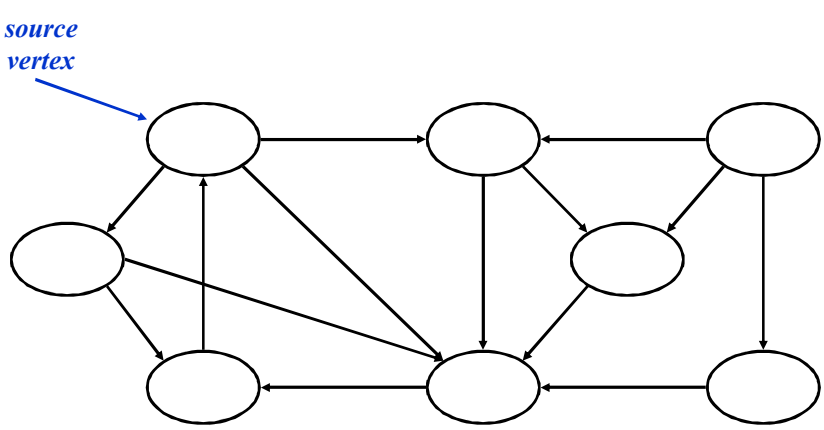

So, running time of DFS = O(V+E)

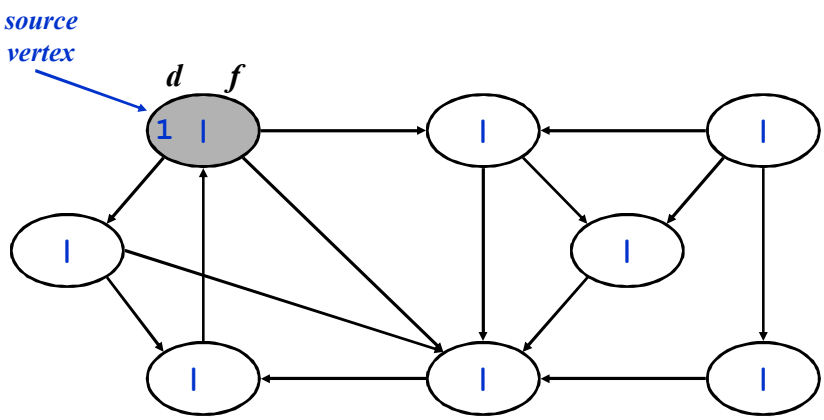
Depth-First Search Analysis

- This running time argument is an informal example of *amortized analysis*
  - "Charge" the exploration of edge to the edge:
    - Each loop in DFS_Visit can be attributed to an edge in the graph
    - Runs once/edge if directed graph, twice if undirected
    - Thus loop will run in $O(E)$ time, algorithm $O(V + E)$
  - Storage requirement is $O(V + E)$, since adjacent list requires $O(V + E)$ storage

DFS Example



DFS Example



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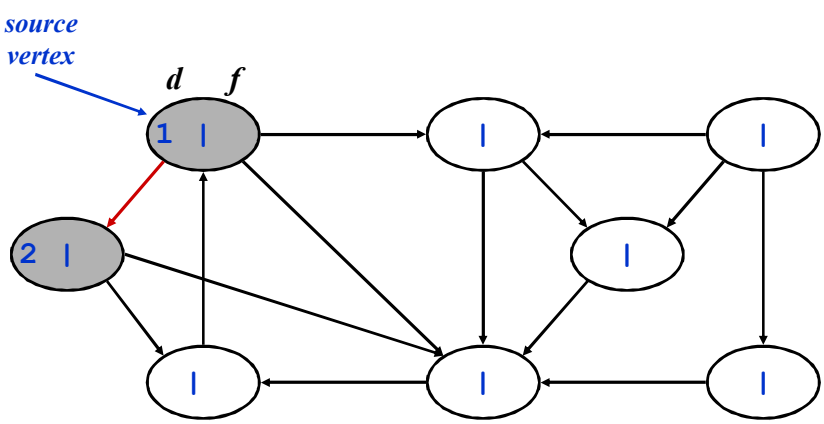
DFS Example



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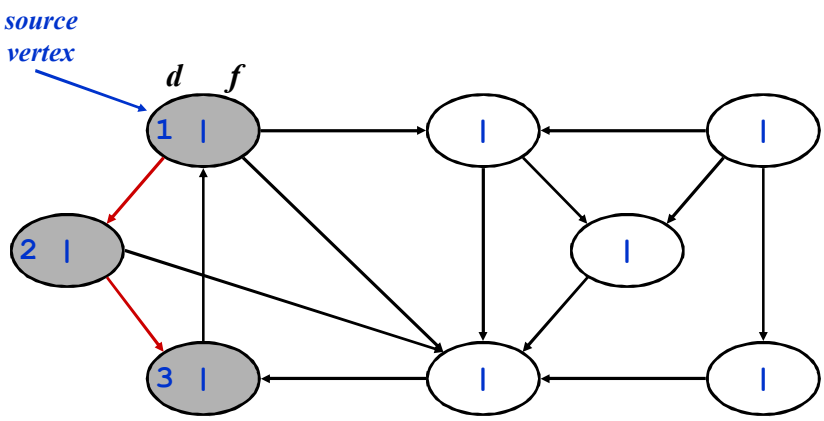
DFS Example



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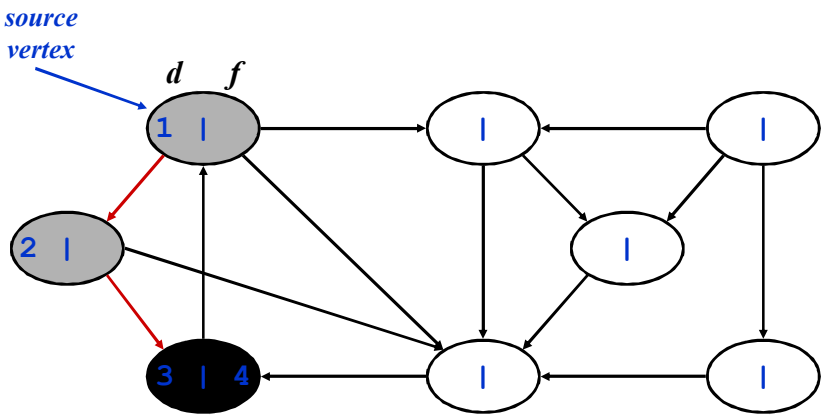
DFS Example



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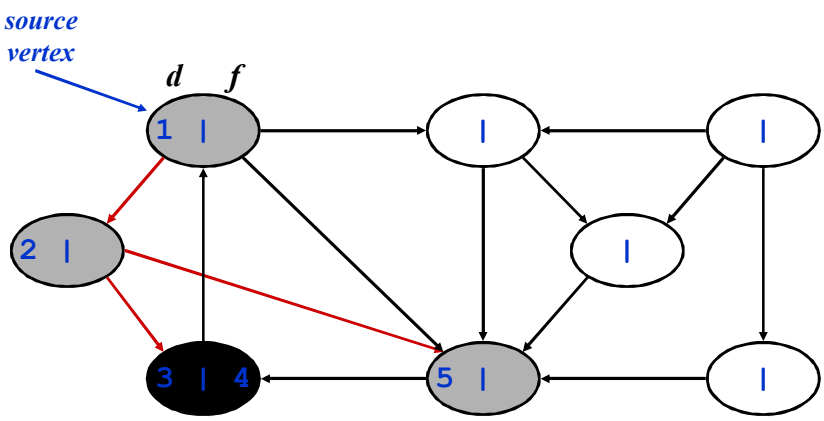
DFS Example



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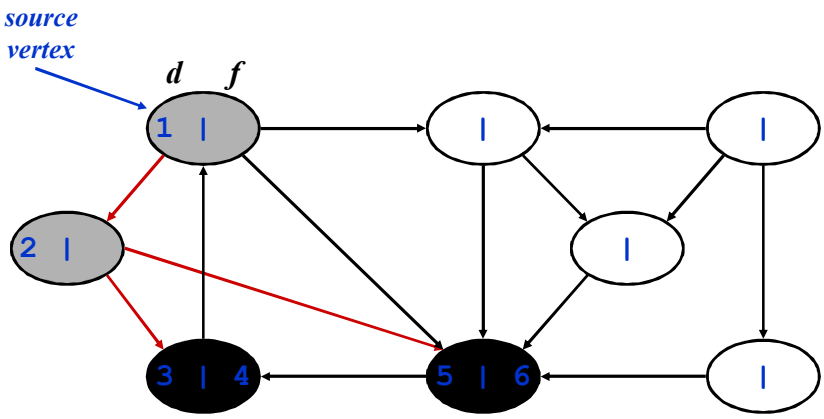
DFS Example



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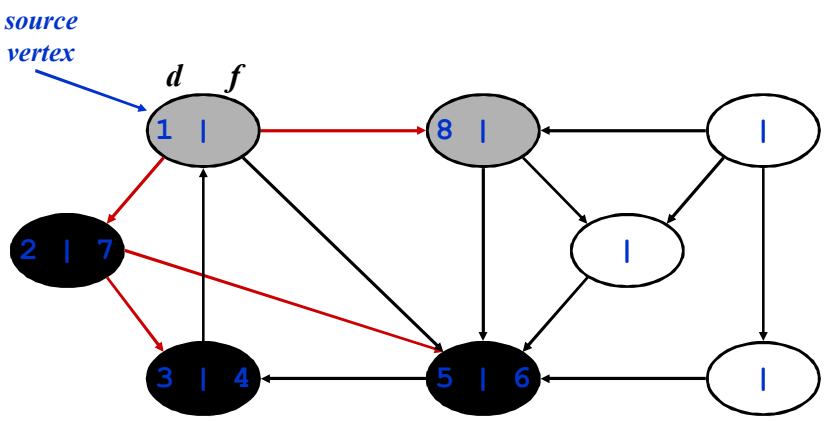
DFS Example



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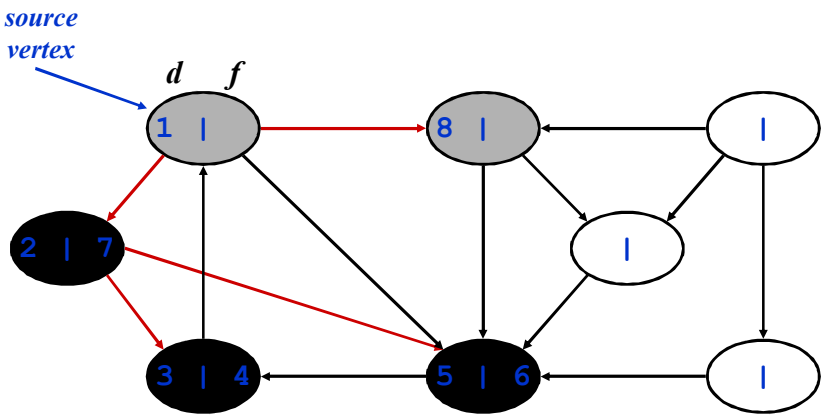
DFS Example



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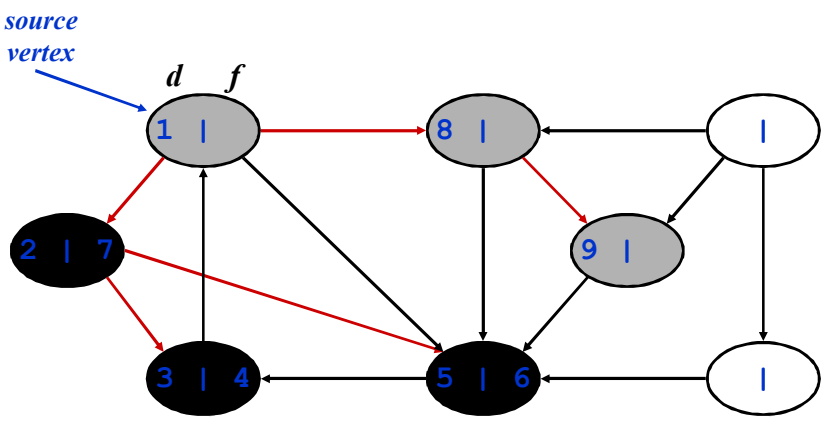
DFS Example



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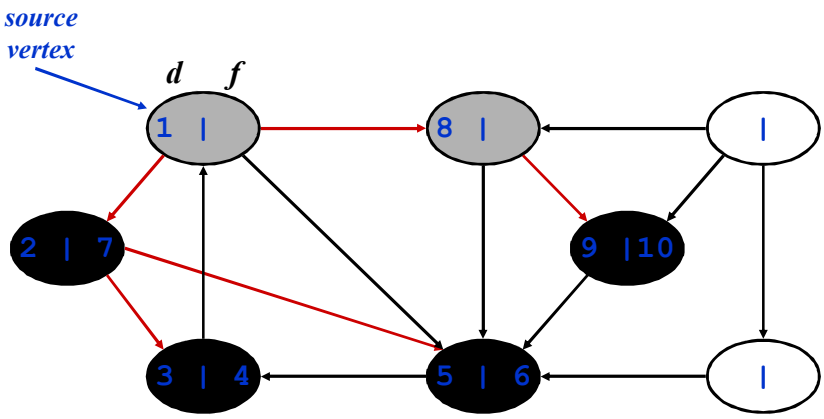
DFS Example



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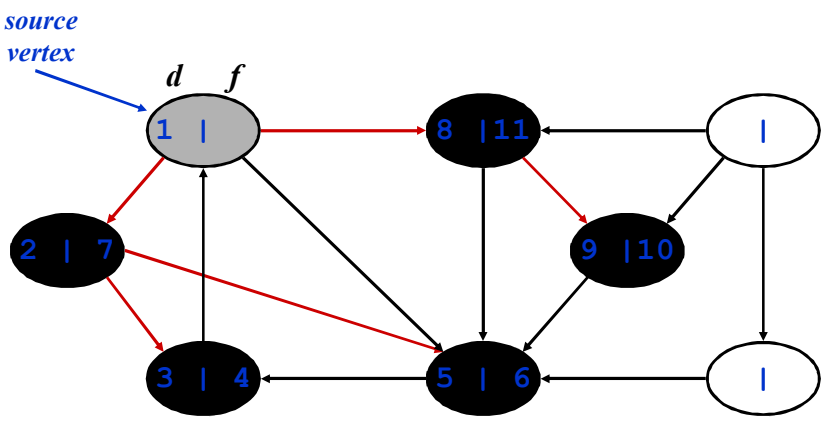
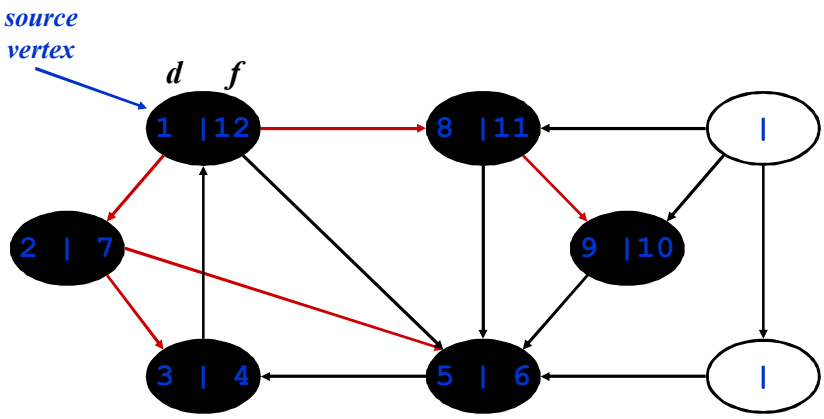
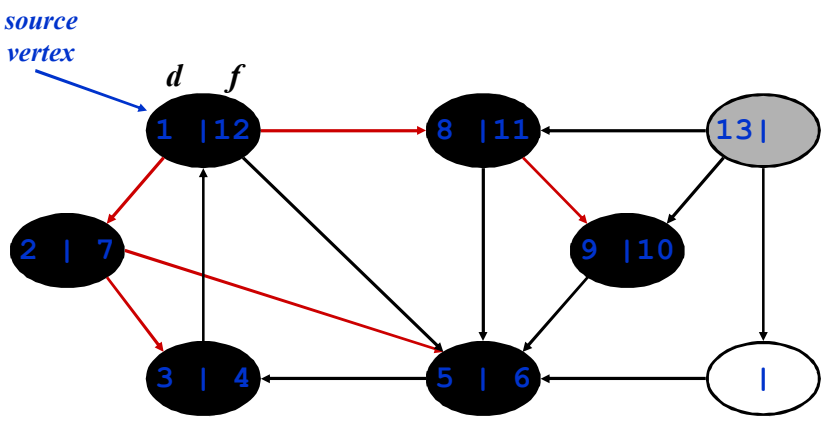
DFS Example



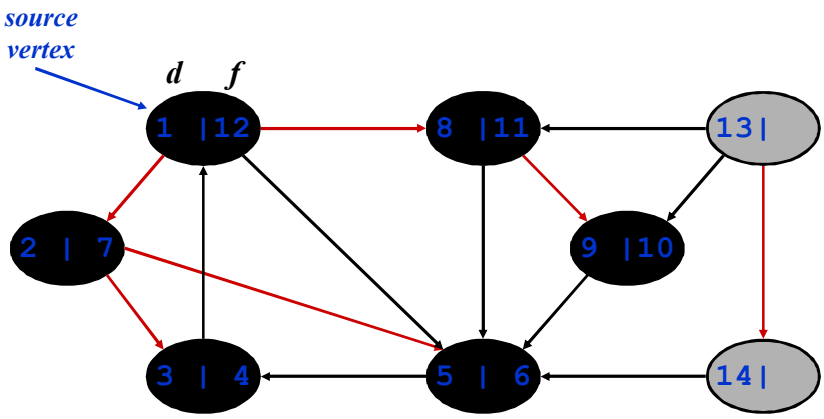
DFS Example



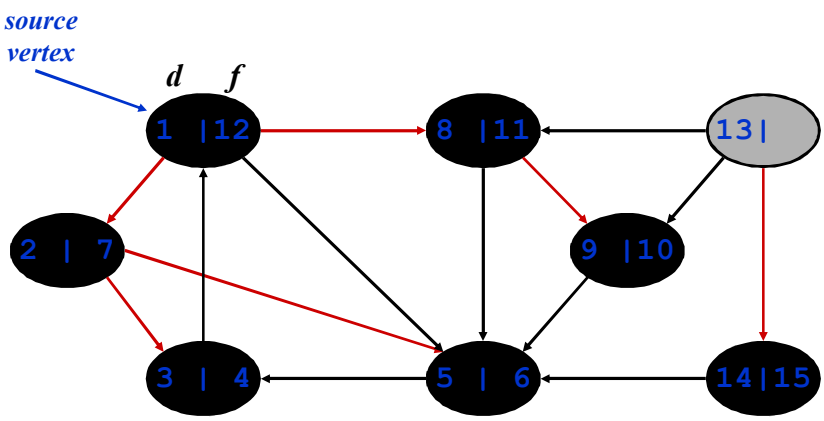
DFS Example



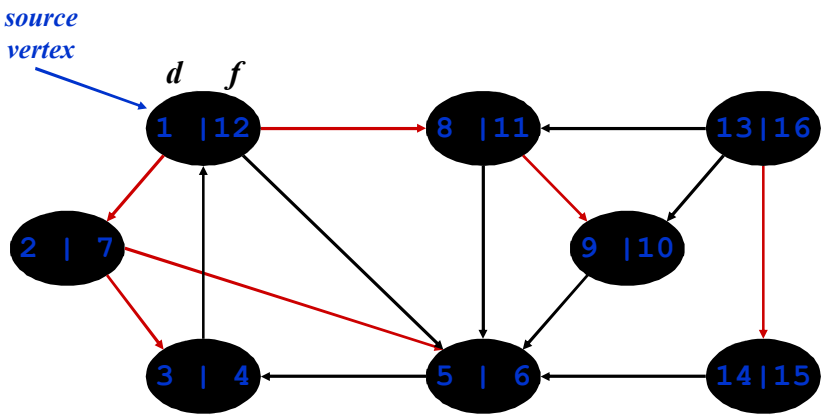
DFS Example



DFS Example



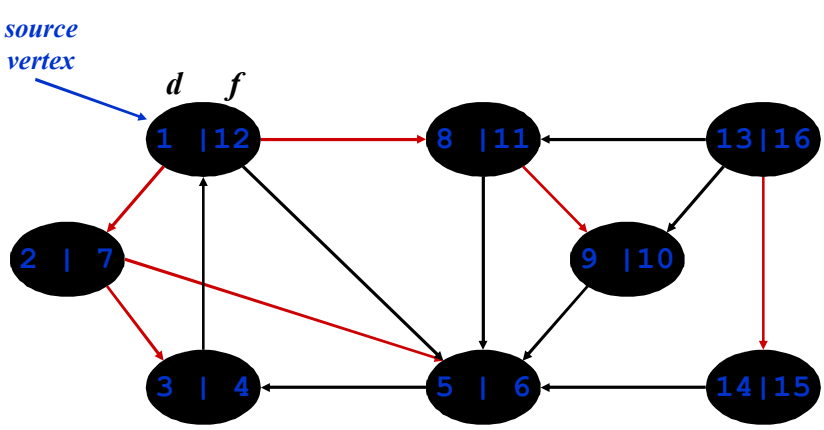
DFS Example



DFS: Kinds of edges

- DFS introduces an important distinction among edges in the original graph:
  - *Tree edge*: encounter new (white) vertex
    - The tree edges form a spanning forest
    - *Can tree edges form cycles? Why or why not?*

DFS: Kinds of edges

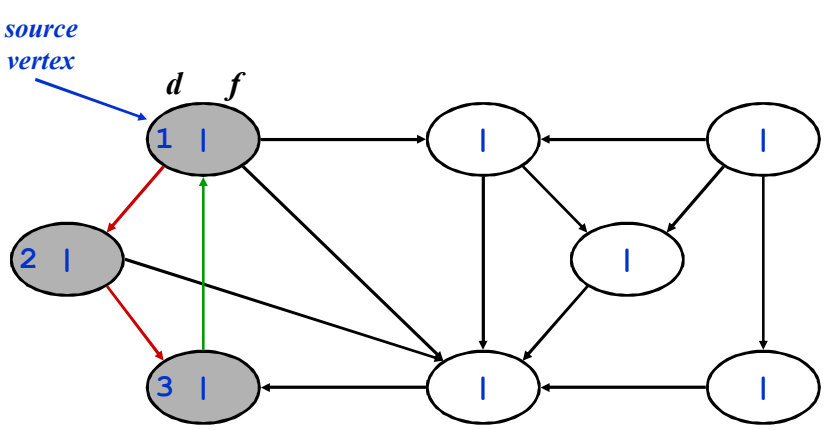


Tree edges

DFS: Kinds of edges

- DFS introduces an important distinction among edges in the original graph:
  - *Tree edge*: encounter new (white) vertex
  - *Back edge*: from descendent to ancestor
    - Encounter a grey vertex (grey to grey)

DFS: Kinds of edges

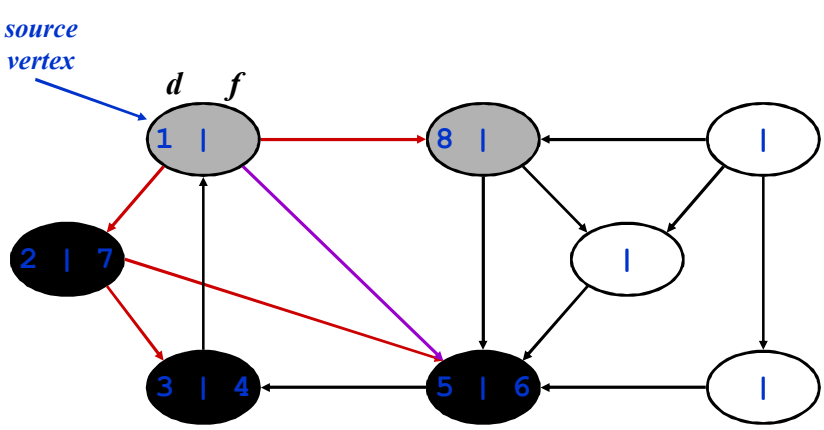


Tree edges *Back edges*

DFS: Kinds of edges

- DFS introduces an important distinction among edges in the original graph:
  - *Tree edge*: encounter new (white) vertex
  - *Back edge*: from descendent to ancestor
  - *Forward edge*: from ancestor to descendent
    - Not a tree edge, though
    - From grey node to black node

DFS: Kinds of edges

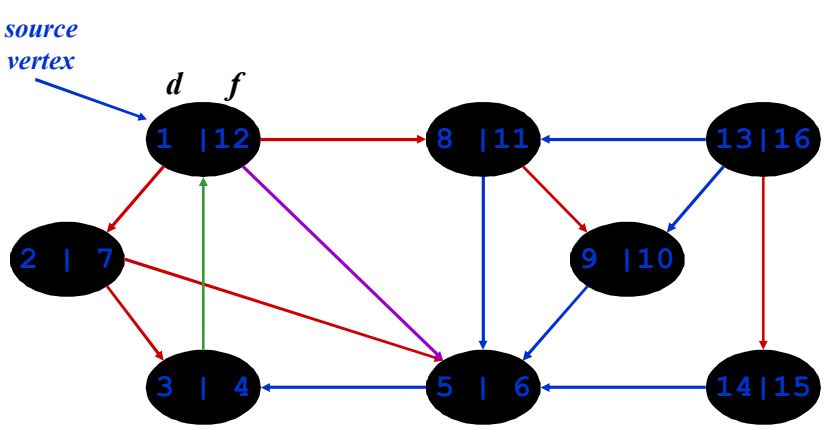


Tree edges *Back edges* *Forward edges*

DFS: Kinds of edges

- DFS introduces an important distinction among edges in the original graph:
  - *Tree edge*: encounter new (white) vertex
  - *Back edge*: from descendent to ancestor
  - *Forward edge*: from ancestor to descendent
  - *Cross edge*: between a tree or subtrees
    - From a grey node to a black node

DFS: Kinds of edges

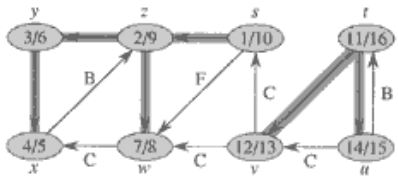


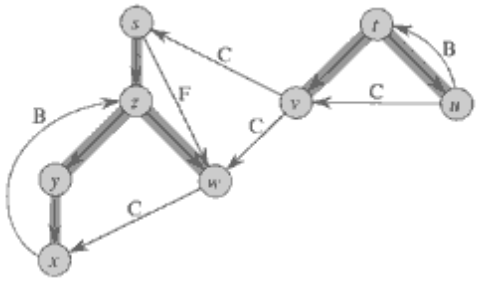
Tree edges *Back edges* *Forward edges* *Cross edges*

DFS: Kinds of edges

- DFS introduces an important distinction among edges in the original graph:
  - *Tree edge*: encounter new (white) vertex
  - *Back edge*: from descendent to ancestor
  - *Forward edge*: from ancestor to descendent
  - *Cross edge*: between a tree or subtrees
- Note: tree and back edges are very important; some algorithms use forward and cross edges

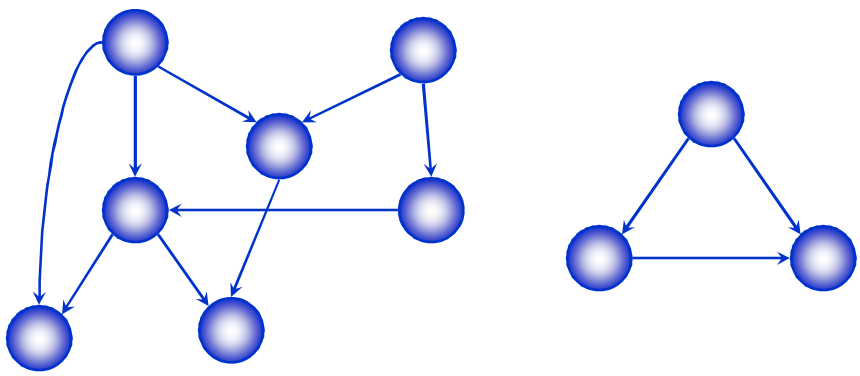
DFS: Kinds of edges





Directed Acyclic Graphs

- A *directed acyclic graph* or *DAG* is a directed graph with no directed cycles



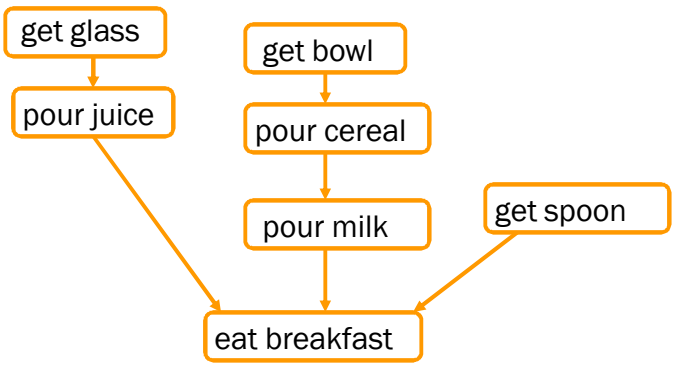
Topological Sort

- A *topological sort* of a DAG is
  - a linear ordering of all vertices of the graph G such that vertex u comes before vertex v if (u, v) is an edge in G.
- DAG indicates precedence among events:
  - events are graph vertices, edge from u to v means event u has precedence over event v
- Real-world example:
  - getting dressed
  - course registration
  - tasks for eating meal

Precedence Example

- Tasks that have to be done to eat breakfast:
  - get glass, pour juice, get bowl, pour cereal, pour milk, get spoon, eat.
- Certain events must happen in a certain order (ex: get bowl before pouring milk)
- For other events, it doesn't matter (ex: get bowl and get spoon)

Precedence Example



Order: glass, juice, bowl, cereal, milk, spoon, eat.

Why Acyclic?

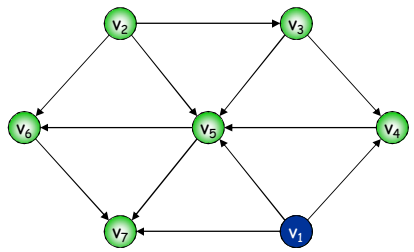
- Why must directed graph by acyclic for the topological sort problem?
- Otherwise, no way to order events linearly without violating a precedence constraint.

Topological Sort Algorithm

```
TOPOLOGICAL-SORT(G)
call DFS(G) to compute finishing times f[v] for each vertex v
as each vertex is finished, insert it onto the front of a linked list
return the linked list of vertices
```

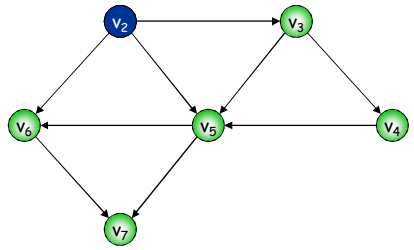
- Time: $O(V + E)$
- Correctness: Want to prove that
  $(u, v) \in E(G) \Rightarrow f(u) > f(v)$

Topological Sort: Using In-degree



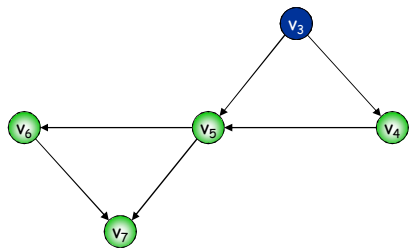
Topological order:

Topological Sort: Using In-degree



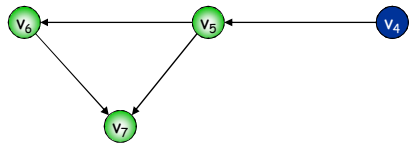
Topological order: v_1

Topological Sort: Using In-degree



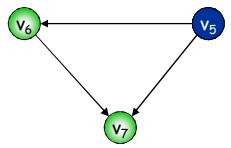
Topological order: v_1, v_2

Topological Sort: Using In-degree



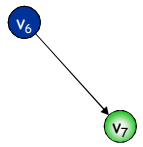
Topological order: v_1, v_2, v_3

Topological Sort: Using In-degree



Topological order: v_1, v_2, v_3, v_4

Topological Sort: Using In-degree

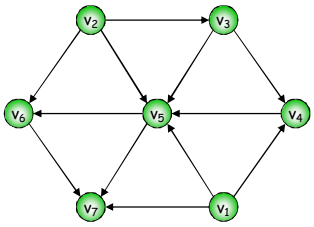


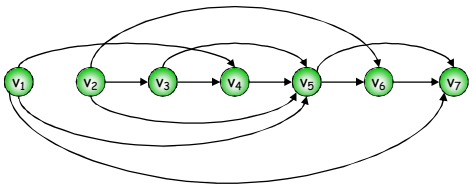
Topological order: v_1, v_2, v_3, v_4, v_5

Topological Sort: Using In-degree

Topological order: $v_1, v_2, v_3, v_4, v_5, v_6$

Topological Sort: Using In-degree





Topological order: $v_1, v_2, v_3, v_4, v_5, v_6, v_7$.

Topological Sort: Using In-degree

Steps for finding the topological ordering of a DAG:

Step-1: Compute in-degree for each of the vertices present in the DAG and initialize the count of visited nodes as 0;

Step-2: Add all **vertices with in-degree equals 0** into a queue

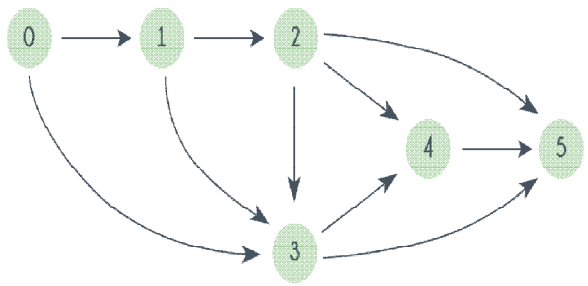
Step-3: Remove a vertex from the queue and then

- Increment count of visited nodes by 1;
- Decrease in-degree by 1 for all its neighboring nodes;
- If in-degree of a neighboring node is reduced to zero, then add it to the queue;

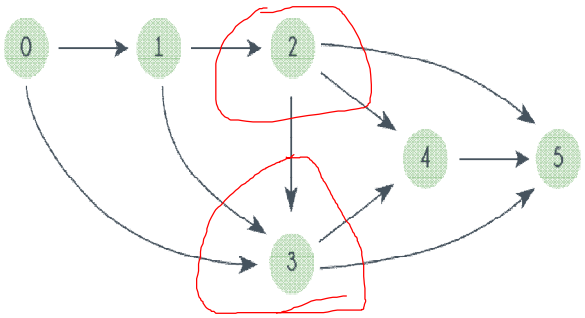
Step 4: Repeat Step 3 until **the queue is empty**;

Step 5: If count of visited nodes is **not** equal to the number of nodes in the graph then the topological sort is not possible for the given graph.

Topological Sort: Using In-degree



Topological Sort: Using In-degree

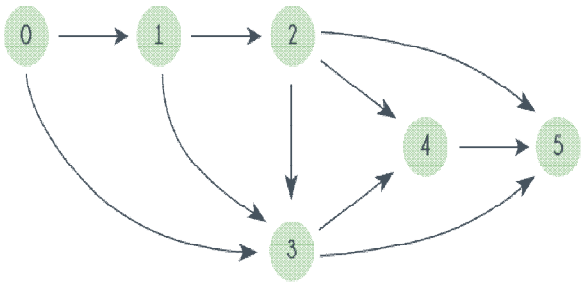


QUEUE : 0

	0	1	2	3	4	5
in_degree	0	1	2	2	2	3

T : --

Topological Sort: Using In-degree

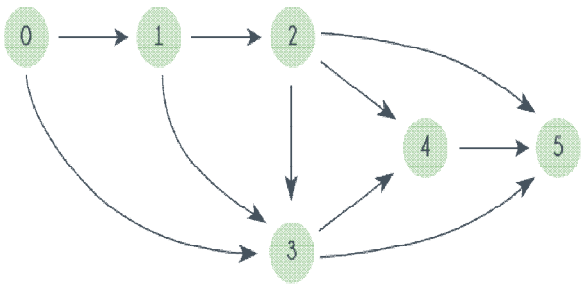


QUEUE : 1

	0	1	2	3	4	5
in_degree	0	0	1	1	2	3

T : 0

Topological Sort: Using In-degree

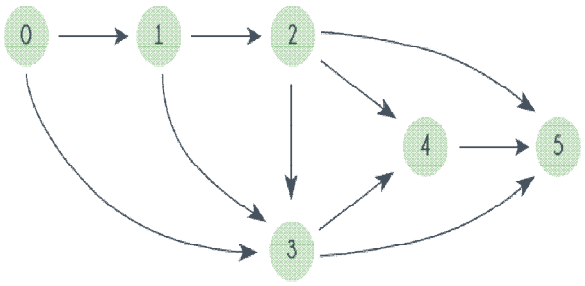


QUEUE : 2

	0	1	2	3	4	5
in_degree	0	0	0	1	1	3

T : 0 1

Topological Sort: Using In-degree

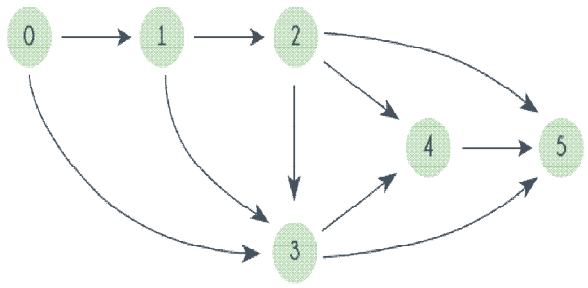


QUEUE : 3

	0	1	2	3	4	5
in_degree	0	0	0	0	1	2

T : 0 1 2

Topological Sort: Using In-degree

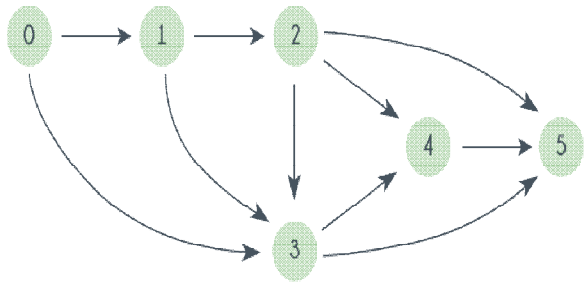


QUEUE : 4

	0	1	2	3	4	5
in_degree	0	0	0	0	0	1

T : 0 1 2 3

Topological Sort: Using In-degree

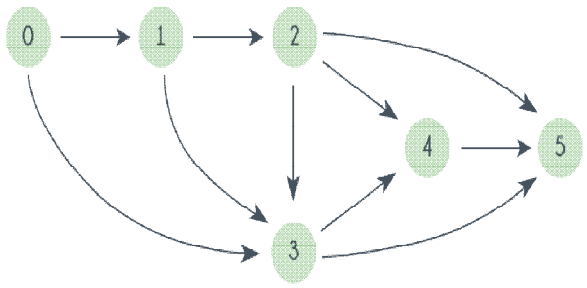


QUEUE : 5

	0	1	2	3	4	5
in_degree	0	0	0	0	0	0

T : 0 1 2 3 4

Topological Sort: Using In-degree



QUEUE : --

	0	1	2	3	4	5
in_degree	0	0	0	0	0	0

T : 0 1 2 3 4 5